

RobotArm-300

Six-Axis Industrial Articulated Arm — Operation, Maintenance & Troubleshooting Manual

MACHINE: RobotArm-300

MODEL: RA-300X

MANUAL VERSION: 3.2

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SECTION: Overview

The RobotArm-300 (model RA-300X) is a six-axis articulated industrial robot arm designed for pick-and-place, welding-fixtured handling, and material transfer applications on mixed assembly lines. It communicates over EtherCAT to a central line controller and reports joint-level diagnostics through its onboard controller unit (CU-300). This manual covers installation checks, routine maintenance, safety procedures, and troubleshooting for the standard fault code range (A0xx series).

This document assumes the reader is a certified line technician familiar with lockout/tagout procedures and basic six-axis robot geometry. It does not cover controller firmware development or custom motion-profile scripting — refer to the RA-300X Programmer's Reference for that scope.

SECTION: Safety Precautions

Before performing any maintenance or fault-clearing procedure on the RobotArm-300, the technician must isolate the arm from motive power and confirm zero-energy state at the controller cabinet. The following precautions apply to all procedures in this manual:

- Engage the physical E-stop at the CU-300 controller before opening any access panel.
- Do not rely on software-disabled motion as a substitute for physical lockout — a communication fault can re-energize joint drives unexpectedly.
- Allow a minimum of 60 seconds after power-down before touching joint 2 and joint 3 servo housings; residual capacitor discharge in the drive units can retain a charge.
- Two-person verification is required before re-enabling motive power after any fault involving joints 1 through 3, due to arm mass and reach.

SECTION: Installation and Initial Setup

Mount the RA-300X base plate to a rigid foundation rated for at least 4x the arm's static load moment. Verify base-plate flatness within 0.1mm across the mounting footprint before torquing the base bolts — an uneven base is the single most common cause of persistent joint-1 vibration faults reported after installation.

Connect the EtherCAT ring cable to the CU-300 controller's IN port, and daisy-chain to the next device on the line via the OUT port. The controller performs an automatic ring-topology check on power-up; a broken ring will surface as a communication fault rather than a joint-specific error.

SECTION: Technical Specifications

The table below lists the core mechanical and electrical specifications for the RA-300X variant covered by this manual.

Parameter	Value
Payload capacity	12 kg
Reach (max)	1,437 mm
Number of axes	6
Repeatability	±0.03 mm
Joint 1 max speed	180°/s
Joint 2 max speed	175°/s
Joint 3 max speed	180°/s
Controller	CU-300, EtherCAT
Rated input power	3-phase, 400V, 3.2 kVA
Operating temperature range	0°C to 45°C
IP rating (arm)	IP65
Mass (arm only)	58 kg

SECTION: Routine Maintenance Schedule

Preventive maintenance intervals are based on single-shift operation (8 hours/day, 5 days/week). Double the frequency for continuous 24-hour operation.

Interval	Task
Daily	Visual inspection of cabling and connector strain relief
Weekly	Check joint 1 base-mount bolt torque
Monthly	Inspect gripper/end-effector wear and calibration
Quarterly	Grease joints 2 and 3 per lubrication chart (Appendix B)
Annually	Full encoder calibration and backlash measurement, all axes

SECTION: Error Code A032 - Joint 2 Encoder Fault

ERROR CODE: A032

A032 indicates the CU-300 controller has lost a valid position signal from the joint 2 absolute encoder for more than 50 milliseconds during motion. This is a hard fault — the controller immediately halts all axis motion when A032 is raised, regardless of current motion profile.

Probable causes:

- Loose or degraded encoder cable connector at the joint 2 servo housing.
- Electromagnetic interference from an adjacent unshielded high-current cable run.
- Encoder battery backup depletion (absolute encoders retain position via battery during power-off; a dead backup battery causes position loss on next power-up).

Corrective action:

1. Engage E-stop and confirm zero-energy state per Safety Precautions.
2. Inspect the joint 2 encoder connector (J2-ENC) for looseness, corrosion, or bent pins.
3. Reseat the connector fully and verify the retention clip engages audibly.
4. If the fault persists, check encoder backup battery voltage at test point TP-4 on the joint 2 servo board; replace if below 2.8V.
5. Clear the fault at the CU-300 HMI (Diagnostics > Clear Fault) and jog joint 2 through its full range at low speed to confirm stable encoder tracking before returning to automatic mode.

SECTION: Error Code A045 - Overtemperature, Joint Drive

ERROR CODE: A045

A045 is raised when any joint servo drive's internal temperature sensor exceeds 85°C. The controller will throttle motion speed automatically at 75°C as an early warning before the hard cutoff at 85°C.

Probable causes:

- Blocked or fouled controller cabinet ventilation fan.
- Ambient temperature at the installation site exceeding the rated 45°C maximum.
- Sustained duty cycle above the rated continuous torque for one or more joints.

Corrective action:

1. Power down and allow the affected drive to cool for at least 15 minutes.
2. Inspect and clean the CU-300 cabinet intake and exhaust fans.
3. Confirm ambient site temperature is within the rated operating range.
4. Review the motion program for excessive duty cycle on the affected joint and reduce cycle rate if the fault recurs after cooling.

SECTION: Error Code A210 - Communication Ring Fault

ERROR CODE: A210

A210 indicates the EtherCAT ring topology check has failed — the controller cannot confirm a closed loop between its IN and OUT ports. This typically appears immediately after power-up or after any cabling work on the line.

Probable causes:

- A disconnected or damaged EtherCAT cable somewhere in the ring.
- A downstream device on the ring powered off or in a fault state that breaks the chain.
- Incorrect port wiring (IN/OUT reversed) at installation.

Corrective action:

1. Check that all devices on the EtherCAT ring are powered on.
2. Inspect cabling at each ring junction for damage or loose connectors, starting from the CU-300 OUT port and working downstream.
3. Confirm IN/OUT port wiring matches the line topology diagram (Appendix A).
4. If the fault clears on isolating a single downstream device, that device's own controller and cabling require separate inspection.

SECTION: Gripper and End-Effector Calibration

The RA-300X ships with a standard two-finger pneumatic gripper mount at the joint 6 flange. Calibration should be verified monthly per the maintenance schedule, or immediately after any end-effector swap.

Access the calibration routine via CU-300 HMI: Tools > End Effector > Calibrate. The routine requires the gripper to be empty and will move joint 6 through a fixed reference sequence. Do not interrupt the sequence once started — an interrupted calibration leaves the tool-center-point offset in an undefined state and the routine must be restarted from the beginning.

SECTION: Storage and Long-Term Shutdown

For shutdowns exceeding 30 days, move the arm to its designated storage pose (all joints at zero position) before removing power, to avoid gravity-induced strain on the joint 2 and joint 3 brake assemblies. Disconnect the encoder backup battery pack if storage will exceed 180 days, and note the disconnect date on the maintenance log — batteries left connected during extended storage typically deplete within 9-12 months and will require a full encoder recalibration on restart.

SECTION: Motion Programming Basics

The CU-300 controller accepts motion programs via its onboard teach pendant or over the line controller's supervisory interface. Each motion program consists of an ordered list of waypoints, each with a target joint configuration, approach speed, and blend radius. This section covers the basic waypoint model only; full programming syntax is documented in the separate RA-300X Programmer's Reference.

Waypoints are stored as joint-space coordinates (six values, one per axis) rather than cartesian coordinates, which avoids inverse-kinematics ambiguity near singularities. When teaching a new waypoint, jog the arm to the desired pose using the pendant's joint-jog mode, then record the pose with the SAVE POINT command. Always verify a newly taught point with a low-speed dry run before including it in a production motion program.

Blend radius controls how sharply the arm transitions between consecutive waypoints. A blend radius of zero forces the arm to fully stop at each point (useful near fixtures or operators); larger blend radii produce smoother, faster motion but require more clearance around the path.

SECTION: Appendix A - EtherCAT Ring Topology

The RA-300X CU-300 controller must occupy a fixed position in the line's EtherCAT ring as defined by the line topology diagram provided at commissioning. In the standard three-cell layout, the CU-300 sits between the upstream conveyor controller and the downstream press controller, with its IN port wired to the upstream device's OUT port and its OUT port wired to the downstream device's IN port.

Reversing IN and OUT at any single device breaks the ring's expected sequence numbering and will surface as an A210 communication fault at the next power-up, even though each individual cable connection may appear physically intact. When troubleshooting A210, always confirm port orientation before assuming a cable fault.

Position in ring	Device	IN from	OUT to
1	Conveyor Controller	Line Master	RobotArm-300 (CU-300)
2	RobotArm-300 (CU-300)	Conveyor Controller	Hydraulic Press Controller
3	Hydraulic Press Controller	RobotArm-300 (CU-300)	Line Master

SECTION: Appendix B - Lubrication Chart

Use only the specified grease grade for each joint. Mixing incompatible greases can cause seal degradation and premature bearing wear.

Joint	Grease Grade	Quantity	Interval
Joint 1 (base)	NLGI 2 lithium complex	15 g	Quarterly
Joint 2 (shoulder)	NLGI 2 lithium complex	20 g	Quarterly
Joint 3 (elbow)	NLGI 2 lithium complex	12 g	Quarterly
Joint 4 (wrist roll)	NLGI 1 synthetic	5 g	Semi-annual
Joint 5 (wrist bend)	NLGI 1 synthetic	5 g	Semi-annual
Joint 6 (tool flange)	NLGI 1 synthetic	3 g	Semi-annual

SECTION: Troubleshooting Flow for Unclassified Faults

If the CU-300 HMI displays a fault code not listed in this manual's error code sections, do not attempt to clear it and resume automatic operation. Record the exact fault code, the HMI's fault description text, and the operation in progress at the time of the fault, then follow the escalation path in the Contact and Support section.

General-purpose fault codes in the A9xx range are reserved for controller firmware diagnostics and are not covered by this operator manual. These require service-level access to the CU-300 diagnostic port and should not be cleared by line personnel.

SECTION: Glossary

Absolute encoder — A position sensor that retains its exact joint position across power cycles, using a small backup battery, as opposed to an incremental encoder which must re-home on every power-up.

Blend radius — The distance before a waypoint at which the arm begins transitioning toward the next waypoint, producing smoother continuous motion.

Tool center point (TCP) — The reference point on the end-effector used for all motion programming, defined relative to the joint 6 flange.

Zero-energy state — A confirmed condition in which the arm has no stored electrical, pneumatic, or gravitational potential energy capable of causing unexpected motion.

SECTION: Operator Control Panel Reference

The CU-300 front panel provides a fixed set of physical controls in addition to the touch HMI, intended for quick status checks and emergency actions without navigating the menu system.

Control	Function
E-STOP (red mushroom)	Immediate hard stop of all axis motion and drive power
STATUS LED (green/amber/red)	Green: ready. Amber: fault, recoverable. Red: fault, service required
RESET	Clears a recoverable fault after corrective action; does not override a red-status fault
MODE SELECT	Toggles between Manual (teach pendant control) and Auto (line-controller control)

SECTION: Environmental and Site Requirements

Beyond the temperature and power specifications listed earlier in this manual, the installation site should meet the following conditions to ensure reliable long-term operation of the RA-300X and CU-300 controller.

- Relative humidity: 20% to 80%, non-condensing.
- Vibration isolation: the mounting foundation should limit ambient vibration transmitted to the base plate to below 0.5G to avoid nuisance encoder faults.
- Ambient electromagnetic environment: maintain at least 300mm separation between the CU-300 controller cabinet and any variable-frequency drive (VFD) not sharing a common ground reference.
- Compressed air supply for the gripper: clean, dry, 5.5 to 7 bar, filtered to 5 microns.

SECTION: Warranty and Service Terms

The RA-300X arm and CU-300 controller are covered under standard terms for parts and labor from the date of commissioning, provided the maintenance schedule in this manual is followed and documented in the unit's service log. Warranty coverage does not extend to damage resulting from operation outside the environmental and electrical specifications listed in this manual, or from unauthorized modification of joint servo assemblies.

Routine consumables — gripper pads, joint grease, and encoder backup batteries — are excluded from warranty coverage as normal wear items.

SECTION: Contact and Support

For fault conditions not covered in this manual, or for repeated faults that persist after following the corrective actions above, escalate to line engineering with the fault code, timestamp, and a description of the operation in progress when the fault occurred. Do not attempt undocumented repair procedures on joint servo assemblies — refer these to certified service personnel.